1 Supplementary Error Analysis

This supplementary appendix provides additional error-distribution analyses omitted from the main submission due to page limitations. These analyses complement the results reported in the main paper and provide analyses of model failure patterns across all evaluated experimental settings.

1.1 Output Prediction Errors

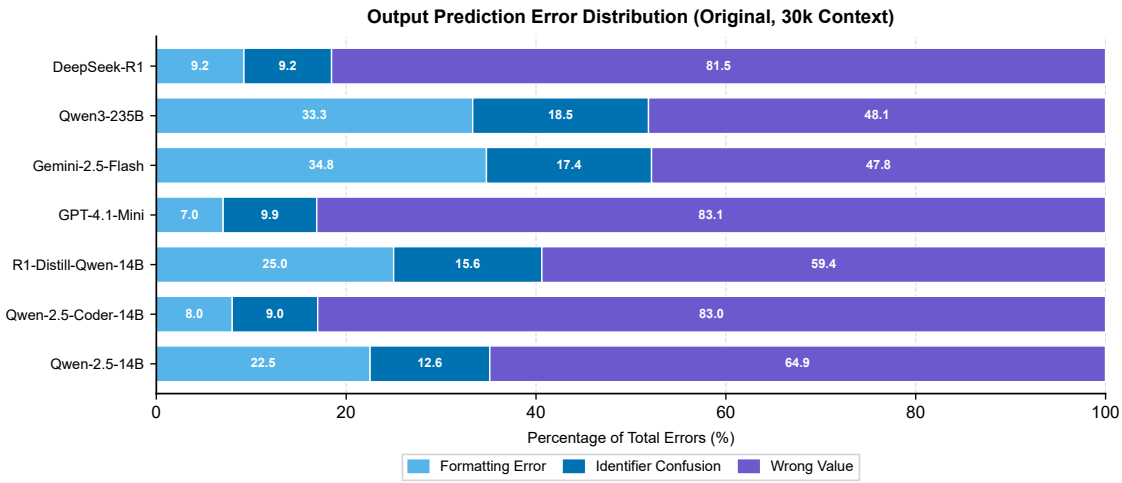

Fig. 1. Output Prediction error distribution on **Original** data under **30k** retrieval context. Wrong Value errors increase for weaker models compared with the corresponding 10k setting reported in the main paper, suggesting increased sensitivity to retrieval noise under longer contexts.

Compared with the 10k retrieval setting reported in the main paper, extending the retrieval context to 30k tokens does not consistently improve performance. Instead, several weaker models exhibit a noticeable increase in Wrong Value errors, indicating that additional retrieved context may introduce distracting information that interferes with accurate execution reasoning. In contrast, stronger models such as GPT-4.1-Mini and DeepSeek-R1 remain comparatively stable, suggesting better robustness to noisy long-context retrieval.

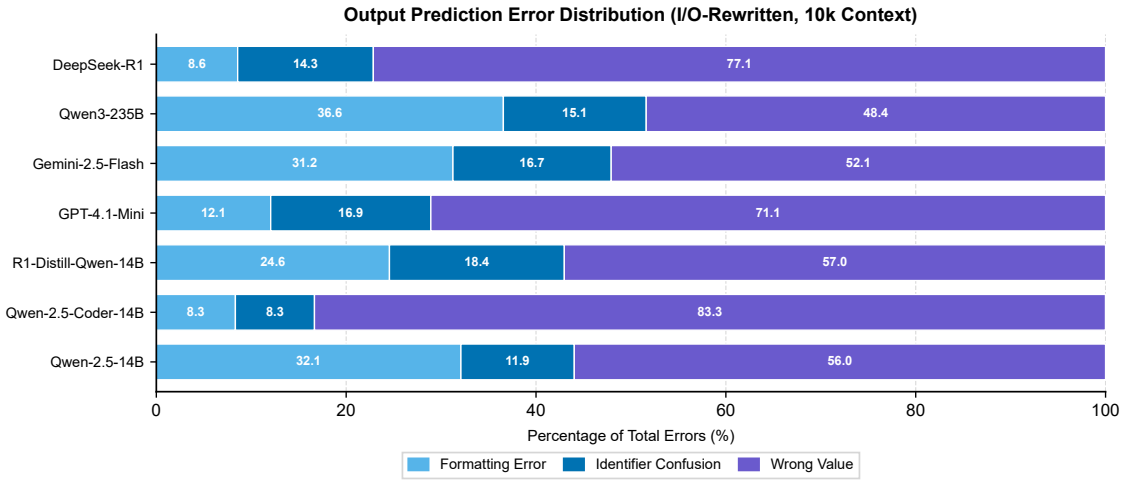

Fig. 2. Output Prediction error distribution on **I/O-Rewritten** data under **10k** Oracle context. Several models exhibit increased Formatting and Identifier Confusion errors relative to the original-data setting, indicating reduced robustness under rewritten but semantically equivalent inputs.

Under rewritten but semantically equivalent inputs, multiple models show shifts in error composition, particularly increases in Formatting Error and Identifier Confusion. This trend further supports the observation in the main paper that model performance may partially rely on memorized value patterns rather than purely execution-based reasoning.

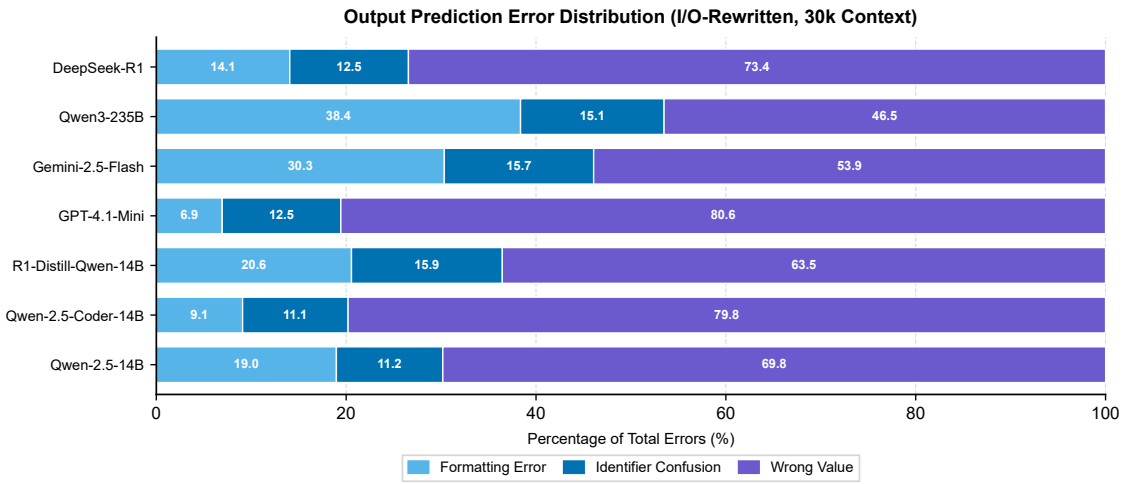

Fig. 3. Output Prediction error distribution on **I/O-Rewritten** data under **30k** retrieval context. Combining rewritten inputs with longer noisy context amplifies the degradation observed in both rewritten and long-context settings.

When rewritten inputs are combined with longer noisy retrieval contexts, the degradation becomes more pronounced across most evaluated models. Wrong Value remains the dominant failure category, indicating that cross-file execution reasoning remains the principal bottleneck even when retrieval coverage is increased.

Overall, across all Output Prediction settings, Wrong Value consistently constitutes the majority of errors. This suggests that the primary limitation of current models lies not in output formatting or identifier selection, but in accurately simulating repository-level execution and propagating state across multiple files.

1.2 Call Chain Prediction Errors

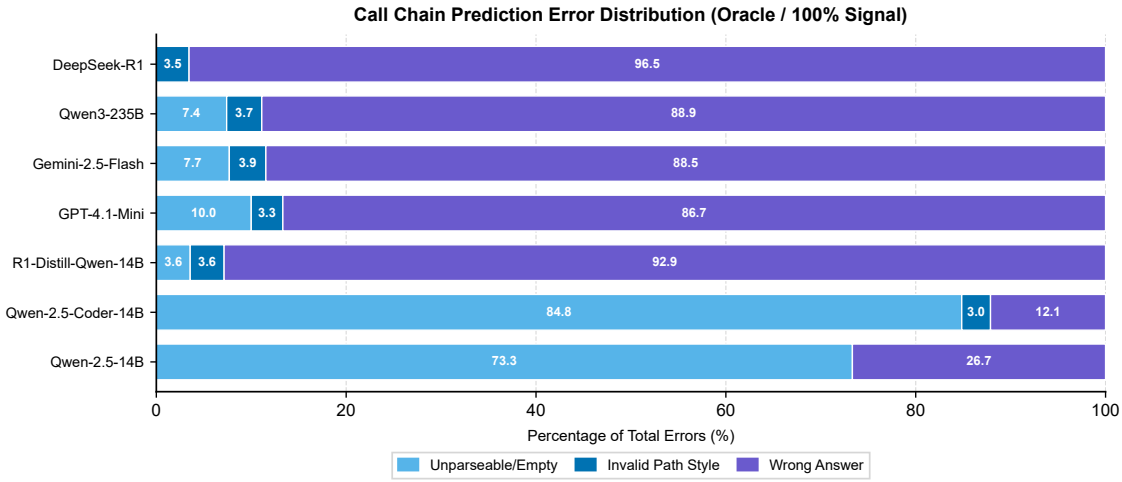

Fig. 4. Call Chain Prediction error distribution under the **Oracle (100% signal)** setting. Even when only ground-truth files are provided, larger models remain primarily limited by incorrect dependency reconstruction rather than formatting failures.

Even under oracle context where all provided files belong to the ground-truth call chain, most capable models predominantly fail by producing incorrect call chains rather than malformed outputs. This indicates that multi-hop dependency reasoning remains challenging even when retrieval noise is completely removed.

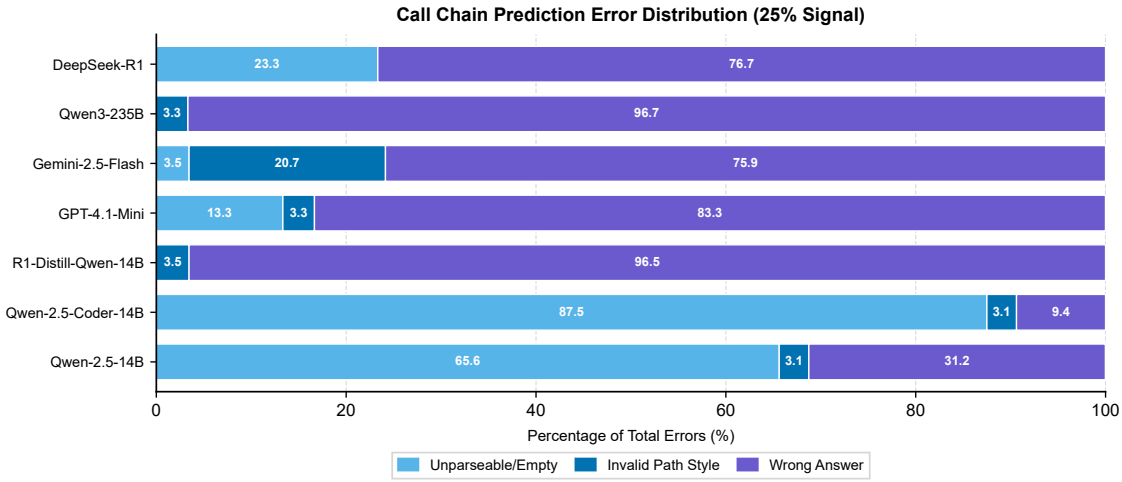

Fig. 5. Call Chain Prediction error distribution under the **25% signal** setting. Distractor noise primarily increases output-formatting failures for weaker models, while stronger models remain dominated by dependency-tracing errors.

When distractor noise is introduced, weaker models show a clear increase in Unparseable/Empty outputs, suggesting that noisy repository context substantially harms their output stability and parsing reliability. In contrast, stronger models remain dominated by Wrong Answer errors, implying that their primary limitation lies in dependency reconstruction rather than noise robustness.

This observation is consistent with the main paper's finding that current LLMs exhibit relatively high precision but limited recall in dependency reconstruction, indicating difficulty in fully recovering multi-hop execution paths.